INCH POUND

MIL-C-3098/118B
24 September 1992
SUPERSEDING
MIL-C-3098/118A
9 May 1984

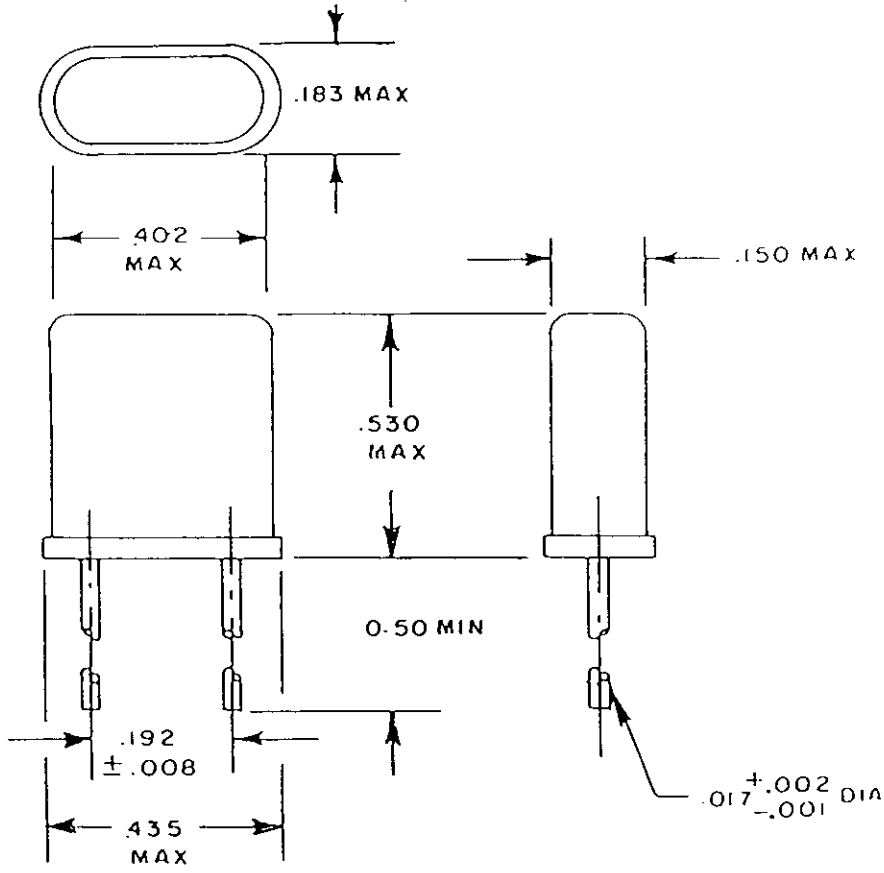
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR139/U

This specification is approved for use by all Department and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 20 to 22 MHz; fundamental; noncontrolled; antiresonance.




| Inches | mm |
|---|---|
| .001 | 0.03 |
| .002 | 0.05 |
| .008 | 0.20 |
| .017 | 0.43 |
| .150 | 3.81 |
| .183 | 4.65 |
| .192 | 4.88 |
| .402 | 10.21 |
| .435 | 11.05 |
| .50 | 12.7 |
| .530 | 13.46 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR139/U.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-18/U or HC-43/U (MIL-H-10056/9).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 20 to 22 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level: 2.5 mW maximum.

Capacitance, shunt: 12.0 pF, maximum.

Antiresonance load capacitance: 30.0 ±0.5 pF.

Equivalent resistance: 40 ohms, maximum.

Shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

NOTE: These crystal units are for use in the RTA-45A equipment. They were previously fabricated using Air Force Technical Order 12-1-20 or Bendix Corporation specification L688450-11 through -20.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0672-02)



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